

Oppositions to the Development of Nuclear Power in Andhra Pradesh and Telangana

7

Varigonda Kesava Chandra

The Government of India is pursuing several nuclear power projects and uranium mining projects simultaneously in the states of Andhra Pradesh and Telangana. These two states were newly formed in 2014 from the erstwhile single state of Andhra Pradesh. Three key projects include: in Andhra Pradesh, the proposed nuclear power plant at Kovvada in Srikakulam district, and the uranium mine and the uranium purification plant at Tummalapalle in Kadapa district; and in Telangana, the uranium mine and purification plant at Lambapur-Peddagattu, in Nalgonda district. The Kovvada nuclear power plant is meant to be a commercial power-generating plant deemed among the largest in terms of operating capacity in the country. The uranium mines in Lambapur-Peddagattu and Tummalapalle would provide the essential fuel for nuclear power development. The government of India along with the state governments of Andhra Pradesh and Telangana, and India's nuclear establishment, have faced considerable opposition from society against these projects. Based on the field research and interviews carried out between 2016 and 2017 in Srikakulam district, Visakhapatnam and Hyderabad, as well as on secondary and archival research, this chapter details social opposition to the Indian state's nuclear power projects in Andhra Pradesh and Telangana.

INTRODUCTION

Since the mid-2000s, the southern Indian states of Andhra Pradesh and Telangana, which till 2014 constituted the single state of Andhra Pradesh, have been selected for major civil nuclear energy projects by the Indian state and India's nuclear establishment. The nuclear establishment constitutes the major technocratic and scientific wings responsible for the formulation and implementation of civil nuclear policy, such as the Atomic Energy Commission (AEC), the Department of Atomic Energy (DAE) and the Nuclear Power Corporation of India Limited (NPCIL). These projects include the nuclear power plant in Kovvada, as well as the uranium mining projects in Tummalapalle, Lambapur-Peddagattu, and Nallamala. The projects were conceptualized in the late-2000s. By the late-2010s, two other projects were also initiated by the Indian government: an additional uranium mine in the Nallamala forests that cut across Andhra Pradesh and Telangana.

The nuclear power plant at Kovvada in the Srikakulam district of Andhra Pradesh gave rise to considerable opposition from the surrounding population since the proposal for its construction was first mooted in the late 2000s. From 2009 to mid-2010s, the residents of the Kovvada region, supported by the surrounding populace, urban-based anti-nuclear activists, and socio-political parties, engaged in a series of collective actions against the decision of the Indian government and the Andhra Pradesh state government to commission a nuclear power project. The opposition movement eventually declined. The state government is, as of 2020, in the process of finalizing land acquisition for the project. However, the project itself remains in abeyance. It is yet to begin construction.

The proposed uranium mine and purification plant at Tummalapalle in the YSR (previously named as Kadapa) district of Andhra Pradesh were also announced in the late 2000s. By 2012, the purification plant had been constructed, and the operation of the mine and purification plant had begun. However, reports of adverse socio-environmental impact from the mines led to considerable apprehension among the surrounding communities, even as the government announced a

further expansion of the mining project. In contrast, the proposed uranium mine and purification plant at Lambapur-Peddagattu, in Telangana's Nalgonda district, did not proceed beyond the exploration stage. Concerted protests from the local populace, supported by civil society organizations as well as major political parties, led to the project's indefinite suspension. As mentioned before, in the mid-2000s, an additional uranium mining project was proposed in a region that fell within the bounds of the Nallamala forests. This too was met with large-scale protests. From 2017 onwards, protests emerged from the Chenchu tribal community residing within the Nallamala. These protests were accentuated through a concerted campaign using social media by civil society groups. Eventually, the campaign to 'save Nallamala' became a *cause celebre* in Andhra Pradesh and Telangana. In 2019, the state and central governments suspended the Nallamala uranium mining project.

CONCEPTUALIZING THE KOVVADA NUCLEAR POWER PROJECT

A nuclear plant in Kovvada, Andhra Pradesh was first mooted in the late 1980s, under then chief minister N. T. Rama Rao of the Telugu Desam Party (TDP). The project, which was slated to be an Indo-Soviet collaboration, was shelved following the disintegration of the Soviet Union in 1991. In the mid-2000s, a possible nuclear power project in Kovvada was briefly mooted again. The state government was then helmed by the Congress party, under chief minister Y. S. Rajasekhara Reddy. In the late 2000s, the nuclear project in Kovvada was revived.

The Kovvada nuclear power project, as envisaged in the late 2000s, featured six nuclear reactors with a total capacity of nearly 10,000 MW, along with an extensive township with the capacity to house 6,000 employees and their families. The project would, if operationalized, possess the largest capacity of any nuclear power project in the country. The Kovvada region, within which the project is to emerge, comprised five villages, Jeerukovvada, Pedda Kovvada, Ramachandrapuram, Tekkali and Gudem, which together amount

to around 500 acres of private land, owned by the local resident communities, and 1,500 acres of government land. The nuclear project was slated to be constructed through a collaboration between NPCIL and the US-based multinational firm, GE-Hitachi. In 2010, the Andhra Pradesh state government, led then by the Congress party which, at the time, also helmed the central government, signalled its intent to proceed with the land acquisition for the nuclear plant (DAE, 1962).¹

OPPOSITION AGAINST THE NUCLEAR PLANT

As soon as the Government of India announced plans in the late-2000s for a nuclear project in the Kovvada region, several urban-based anti-nuclear activists got together to publicize its potentially adverse impact, especially among the local people of the region. These activists, belonging to various NGOs in Visakhapatnam, Srikakulam and Hyderabad, visited the Kovvada region to educate the locals on the adverse impact of the nuclear project and the need to oppose its inception.

The construction of the nuclear power plant would involve depopulating the Kovvada region and relocating its residents. Further, the project could potentially reduce the quantity of fish in the coastal waters. A sizable population in the Kovvada region and its environs were dependent on fishing for their livelihood. Opposition to the nuclear plant, therefore, became firmly rooted among the resident population of the Kovvada region. Furthermore, the deleterious impact of a nuclear power plant on the environment and the potentially harmful impact of radiation on the local populace also caused considerable anxiety in the surrounding region.

Concerted opposition to the project was, therefore, spearheaded by the populace that resided in the Kovvada region. This populace was most at risk of losing their lands and their livelihoods if a nuclear project was to be commissioned in the region. They were supported in their opposition by urban-based anti-nuclear and environmental activists from Hyderabad, Visakhapatnam and Srikakulam. These

independent activists, along with those affiliated with organizations such as Human Rights Forum (HRF) and the nation-wide National Alliance of People's Movements (NAPM) were driven by an anti-nuclear, anti-neoliberal and pro-environment ideology. An umbrella anti-nuclear collective, the Anuvidyut Kendram Vyatireka Porata Udyamam (loosely translated as 'The movement in opposition against nuclear power plant') (AKVPU), was formed at this time.²

The Fukushima-Daiichi nuclear accident, which occurred in Japan in March 2011, brought home to the local people the potentially catastrophic impact of a nuclear power plant, especially in cases of fatal operational flaws, both human and technical. In the ensuing months, reports of mass evacuations of entire villages in the aftermath of the disaster, as well as of possibly irreparable damage to the surrounding environment due to radiation, highlighted the potential for a Fukushima-like disaster in coastal Andhra Pradesh. The incident in Japan gave the burgeoning social opposition in Kovvada greater momentum. It helped spread the opposition to the nuclear project into villages beyond the Kovvada region.

In March 2011, representatives from 150 villages collectively submitted a memorandum to the Mandal (the term 'Mandal' refers to a local government area, part of the district) Revenue Officer citing their opposition to, and demanding a withdrawal of, the plan for the nuclear power plant. In the months following, the villagers submitted further petitions to many officials up to the District Collector, registering their protest against the proposed nuclear project and the resultant land acquisition. In the ensuing months, protests and rallies were held intermittently, often in collaboration with urban-based activists and organizations such as the NAPM and HRF or local socio-political parties, such as the Communist Party of India (Marxist) (CPI[M]), the Centre for Indian Trade Unions (CITU), and the Communist Party of India (Marxist-Leninist) (CPI[ML]). In August 2011, a five-day cycle rally was held in the villages of the Ranasthalam Mandal to educate the local populace on the potential dangers of a nuclear plant, and to shore up support for the opposition (*Dianuke.org*, 2011). Saraswati Kavula, one of the anti-nuclear activists, describes the rally thus:

On day one we were only on cycles as there were only ten yatris. On day two onwards more people joined... Day 3, we rode 8 cycles, 2 motorcycles and 2 auto rickshaws (40 people). Day 4 we had 10 cycles... 1 motorcycle and four auto rickshaws (75 people), Day 5 we had 9 cycles, 1 motorcycle and 3 auto rickshaws (50 people)... The yatris travelled from village to village, disembarked at the beginning of the village and walked through the villages saying anti-nuclear slogans, showing posters to the people, interacting and explaining to them on the ill effects of living near the nuclear plants, distributing pamphlets on the problems from the proposed nuclear power plant.

In April 2012, E. A. S Sarma, a Visakhapatnam-based former bureaucrat-turned-anti-nuclear activist, wrote to then Prime Minister Manmohan Singh, petitioning him to halt the Kovvada nuclear project. Sarma questioned the legality of proceeding with land acquisition when the NPCIL had not yet fulfilled requisite legal obligations, including obtaining environmental clearance from MoEF (*Dianuke.org*, 2012). Notwithstanding the protests by the local populace and urban civil society activists, in November 2012, the Andhra Pradesh state government issued a notification for land acquisition to commence in the Kovvada region. A total population of 7,963 residents were expected to be displaced as a result of this (*India Today*, 2012).

By December 2012, the central and state governments had still not budged. The villagers of the Kovvada region resorted to a novel mode of opposition: a year-long relay hunger strike. Nearly all the villagers in Kovvada, comprising around 5,000 residents, participated in the hunger strike. Groups of villagers, including women, the elderly and children, would take turns to observe the strike daily. Several villagers from the surrounding regions sat with the strikers to express their solidarity. Urban-based activists, including prominent activists such as Medha Patkar of the Narmada Bachao Andolan (NBA), also joined the protesters in solidarity.

The main opposition party in Andhra Pradesh during this period was the Telugu Desam Party (TDP). The local functionaries of the TDP lent material and moral support to the burgeoning anti-nuclear protests. Relations between the local TDP leaders and the residents of the Kovvada region had indeed been quite strong. The Sarpanch

(local elected head) of the Kovvada region and one of the key leaders of the protest movement also contested the Panchayat elections on a TDP ticket, helping cement the party's support for the protest movement. The TDP also lent material support to the relay hunger strike.

A resident of the village described the hunger strike thus:

My friend, the hunger strike is part of the Satyagraha; it has been bequeathed to us by Mahatma Gandhi. It is both our tradition and our right. In this country, if one were to fight for anything, any right, it must only be fought non-violently. Gandhi won our independence from the British in that manner. He has shown us the path. It is in that path that we tread. For one full year, we had carried out the relay hunger strike, continuously without any break, all of us, each one of us in the village. We had displayed our determination, our resolve. And yet, despite all of this, the government hadn't budged. The government hadn't sympathized with us whatsoever. Every resident of this village, all 5,000 of us, took part in the strike, the young, the old, kids, the elderly, men, women, everyone. It was a statement that this land, this village is ours and that we will protect it. We were formed into a team, from every house, every street, and every neighbourhood. And each team would sit in for the hunger strike for a day, on a rotational basis. The plan was created under the auspices of the Sarpanch and our panchayat, as to who should sit in for which day. And in this manner, this relay hunger strike went on for all 365 days of the year (Varigonda, 2020, p. 163).³

The relay hunger strike ended on 17 December 2013, a year after it had begun. However, neither the central nor state government budged in its decision on the inception of the nuclear plant. A resident of the village described the general feeling on the last day of the protest: 'As if this was our last day in our village... as if we would be exiled today... as if all our efforts had been for naught' (Varigonda, 2020, p. 164).

In March 2013, Capt. J. Rama Rao, a Hyderabad-based social activist, filed a public interest litigation (PIL) in the Andhra Pradesh High Court against the acquisition of land in Kovvada. Arguing along similar lines as Sarma, Rama Rao alleged that the land acquisition could only legally be carried out once the environmental clearance had been obtained. In response, the High Court granted an interim stay over land acquisition. However, by September 2013, the High Court ruled in favour of the state and central governments. While directing

the government to adhere to all regulations before the construction of the nuclear plant, the High Court refused to stay the acquisition of land for the nuclear project (*Deccan Chronicle*, 2013).

State's Response: Utilizing a 'Carrot and Stick' Policy

The Indian state and nuclear establishment had not entirely ignored the burgeoning opposition, and the latter's efforts to raise awareness about the dangers of the nuclear project. NPCIL, in partnership with the state government, undertook a concerted public outreach programme among the villagers within and outside the Kovvada region. NPCIL officials interacted with villagers throughout the region, reassuring them of the inherent safety of the plant. Local state functionaries such as the District Collectorate and the Mandal Revenue Office organized various *gram sabhas*, or village councils, in which NPCIL officials interacted with the villagers and talked to them about the safety and security aspects of the project (Srivastava, 2017). As part of the public outreach programme, many of the village representatives were taken on tour to the nuclear plants in Kalpakkam and Kudankulam, in the neighbouring state of Tamil Nadu, to demonstrate their salutary effects on the local people.

Soon it became clear that the Indian state and the nuclear establishment was employing a 'carrot-and-stick' policy towards the residents of the Kovvada region to subjugate the burgeoning opposition. The state government and NPCIL promised attractive compensation packages to the residents of Kovvada in exchange for their land. They promised monetary compensation above the market rate, the construction of a new colony to house the villagers, and assistance with finding new livelihood opportunities. At the same time, the state government had, since at least 2012, enforced an unofficial boycott of the Kovvada region in terms of carrying out any public works. For a populace that was poor and dependent upon essential public works, the lack of cooperation from the state was particularly debilitating. In addition, loans from state-affiliated banks were impossible to acquire, since land that had been earmarked by the state for the acquisition was no longer

accepted as collateral. The situation created by the state government made it clear to the residents of the Kovvada region that they could either negotiate for compensation for the acquired land or continue to fight a lost battle against the nuclear power plant. Having engaged in active opposition against the plant for over five years, many of the residents were also exhausted by this time.

In mid-2014 the Andhra Pradesh state was bifurcated into two states: Telangana and Andhra Pradesh. The same year, the TDP won the State Assembly elections in the newly-created Andhra Pradesh. The erstwhile opposition party was now the party that helmed the government. While the TDP had, at least at the local level, lent its support to the anti-nuclear movement, it changed tack once in power: the TDP-led state government now supported the establishment of a nuclear plant in Kovvada. That same month, the Lok Sabha elections for the Indian parliament took place, with the opposition Bharatiya Janata Party (BJP) emerging victorious. The BJP-led central government under Prime Minister Narendra Modi and the TDP-led state government under Chief Minister Chandrababu Naidu had similar strategies with regard to development. For instance, Chief Minister Naidu's flagship project was the creation of a new capital city (called Amaravati) for the newly-created state of Andhra Pradesh—a massive urbanization and industrialization project.

Such a developmental strategy—which prioritized infrastructure development and industrialization—would, the nuclear establishment hoped, be further dependent upon civil nuclear energy as a key source of power generation (*The Hindu*, 2014). Speaking in support of the Kovvada nuclear project, K. Haribabu, Member of Parliament (MP) for Visakhapatnam for the BJP stated, 'In the long run, hydel power, thermal and gas-based power generation would become unsustainable. In this context, the safe and clean power options we have to look at are solar, wind and nuclear power.... Wind power is highly erratic and undependable. The only viable option we have is nuclear power' (*The Hindu*, 2015). It was, therefore, no surprise when Chief Minister Naidu made it clear that his state government would not oppose central government directives regarding the Kovvada nuclear power project. A plea was made to that effect by the village representatives of the

Kovvada region along with the elected TDP Member of the Legislative Assembly (MLA) of that region, Kala Venkata Rao, to which Naidu reportedly responded by saying, 'If you oppose the NPP [nuclear power project], industrialists will not come to Srikakulam and this will affect development' (*Deccan Chronicle*, 2014).

It was evident that the movement had, by this time, become excessively dependent on the support of local politicians and leaders who were associated with the TDP. The TDP's volte-face, therefore, further contributed to the movement's exhaustion. In mid-2016, the village Panchayat, following a public consultation with the villagers, decided to forgo their opposition and to instead negotiate with the state for adequate compensation for their land.

With the residents of the Kovvada region no longer active participants of the protest movement, the gap was filled by leftist socio-political parties such as CPI (M), CPI (ML), CITU, urban-based activists, and organizations such as HRF, who continued to organize protest rallies. In May 2017, the various organizations, leftist parties, NAPM and prominent urban-based social activists held a meeting in Visakhapatnam with the intent to revive an umbrella organization—another *samiti* (loosely translated as 'committee')—that could present a united front of the opposition. They also attempted to re-induct the villagers of the Kovvada region into the movement again. One activist suggested keeping women at the forefront of the protest movement. Notwithstanding several protest rallies in Visakhapatnam and Srikakulam, an organized, large-scale social resistance did not re-emerge. Without the participation of the villagers of Kovvada, the protests stood weakened.

A Dormant Movement, a Delayed Project

In the 2020 state assembly elections, the incumbent TDP was defeated by the Yuvajana Sramika Rythu Congress Party (YSRCP), a breakaway faction of the Indian National Congress party. The YSRCP is helmed by Y. S. Jaganmohan Reddy, the son of Y. S. Rajasekhara Reddy. In 2020, the YSRCP-led state government, in collaboration with NPCIL, succeeded in acquiring land for the nuclear plant. Successive state governments in Andhra Pradesh, whether led by Congress, TDP or YSRCP,

continued to support the nuclear project. This despite the protracted social protest movement against the nuclear project in Kovvada.

However, as of 2021, the project remains suspended in the planning stages. In the ensuing years, GE-Hitachi dropped out of the project, due in part to disagreements with NPCIL and the Indian government, particularly its laws regarding supplier liability, and in part, to its own precarious financial situation. The project was then offered to another US-based multinational firm, Westinghouse. In early 2017, Westinghouse also dropped out of the project. Its parent company, Toshiba, had just declared bankruptcy. To the multinational firm, the nuclear project at Kovvada was no longer financially viable. NPCIL is yet to decide on a multinational firm with which to collaborate for the construction of the plant. If no multinational firm comes forward, they also have an option to proceed with the plant's construction through indigenous means.

While an anti-nuclear movement on the ground is no longer present, anti-nuclear activists in the region have not yet given up hope. One activist stated that if the construction of the plant commences, the movement would begin anew. It may perhaps no longer be concentrated in Kovvada. As the Kovvada nuclear project is still in abeyance, the anti-nuclear movement also continues to remain dormant.

CONCEPTUALIZING A URANIUM MINING PROJECT: TUMMALAPALLE, LAMBAPUR AND NALLAMALA

Uranium is the primary fuel for nuclear reactors, and India imports most of its uranium. In such a situation, it goes without saying that indigenously extracted and produced uranium would obviate the need to import uranium, and pave the way for energy independence for the country. In the mid-2000s, the Uranium Corporation of India Limited (UCIL) planned to open a large mine along with a purification plant in Tummalapalle. The Tummalapalle region is situated in the Kadapa district (presently renamed as the YSR district) of Andhra Pradesh. In 2007, after obtaining the requisite environmental clearances, the state government proceeded with land acquisition. Over 2,000 acres of government and private land was acquired in

the villages of Tummalapalle, Rachakundapalle, KK Kottaland and Mabbuchintalapalle. A ground-breaking ceremony for the mine and the plant was also conducted that year. In 2011, it was further found that the Tummalapalle region possessed perhaps among the largest uranium deposits in the world—over 1,00,000 tonnes of uranium. By 2012, the construction of the mine and attendant purification plant was completed. The purification plant could treat 3,000 tonnes of uranium ore per day (Subramanian, 2020). The mine and plant commenced their operations in 2012.

In the mid-1990s, uranium reserves had been found in the Lambapur-Peddagattu region, in the Nalgonda district of united Andhra Pradesh (now Telangana). When UCIL planned to establish an open pit and three underground mines along with a purification plant in the region, in the mid-2000s, in addition to private and agricultural land, it also planned to encompass over 1,000 acres of forest land. Conditional environmental clearance was obtained from the MoEF. What was left was environmental clearance from the state government, which would pave the way for the acquisition of land from the local people.

In the mid-2010s, UCIL floated the establishment of yet another uranium mine, this time in the Nallamala forest. Uranium reserves were discovered in the Nallamala forest, which stretched across the Andhra Pradesh and Telangana states, in 2015. In 2017, the Indian government's National Board of Wildlife permitted the Atomic Minerals Directorate (AMD) to explore uranium reserves in the region. The uranium mining project, if realized, would reportedly cover over 20,000 acres of land in the Nallamala. Correspondingly, 20,000 acres of land would be subject to deforestation.

The Socio-environmental Impact of the Uranium Mining Project in Tummalapalle

The acquisition of land for the uranium mine in Tummalapalle, in the Kadapa (later YSR) district of Andhra Pradesh, in the late-2000s, proceeded with purportedly little opposition. N.M. Bahl, a director of UCIL stated, 'There is not a single grievance with regard to

compensation... More and more people are willing to give their land to UCIL in recognition of its corporate social responsibility activities and the benefits extended by UCIL to genuine, native landowners' (Subramanian, 2020). However, this was only part of the story. The public hearing for the Tummalapalle mine was held on 10 September 2006. One activist recounted the strong police presence at the hearing, as well as how several anti-nuclear activists were beaten up by the police; another activist who was present there had even received death threats (Joseph, 2008).

Tummalapalle falls under the constituency of the influential Yedugiri family. Members of the Yedugiri family have been elected to the legislative and parliamentary constituency seats of the region since the 1990s. The family has also produced a former chief minister, Y. S. Rajasekhara Reddy (2004–9) and the current chief minister, Y. S. Jaganmohan Reddy (2019–). Y. S. Rajasekhara Reddy had accorded his enthusiastic support for the mine when as far back as 2006, he announced that the first uranium mine in Andhra Pradesh may be in his constituency (Ramana, 2013). It was during his tenure that land acquisition for the mine was carried out. In 2012, the Tummalapalle mine and the purification plant were commissioned. Since then, the uranium project continues to remain in operation.

In 2016, researchers from the nearby colleges found that the soil and water samples in the Tummalapalle region had deteriorated. The research stated, 'The increased levels of barium, arsenic, cobalt, chromium, copper, molybdenum, lead, vanadium and yttrium are a major concern for the suitability in agricultural and other land management practices' (*The Times of India*, 2016). Increasing incidences of diseases and ailments, including among children, along with higher incidences of crop failure, deaths of livestock, and a steep decline in groundwater continued to be reported in the region (Shree and Serupally, 2018).

These aspects led to protests by the local people and anti-nuclear activists against UCIL. In one instance, residents of seven affected villages in the Tummalapalle region staged a protest by blocking UCIL vehicles; the issue was about providing purified water, as the water in the area was suspected of having become contaminated.

In response to the protests, the Andhra Pradesh state government initiated an inquiry into the mine's operations and its impact on the surrounding environment. The report submitted by the Andhra Pradesh Pollution Control Board (APPCB) stated, 'Uranium concentration in the borewell is in much higher limits, ranging from 690 ppb (parts per million) to 4,000 ppb, whereas the permissible limit is 60 ppb' (*The Times of India*, 2020). One of the key reasons imputed for the higher uranium concentration included UCIL taking inadequate precautionary measures. As the APPCB reportedly stated, 'UCIL failed to line the tailing pond with 250-micron polyethylene layer as directed in consent for operation order of the APPCB.' While admitting to higher concentrations of uranium in the groundwater, controversially, the APPCB did not hold the uranium project squarely responsible for it. Instead, it issued guidelines to UCIL to comply with existing environmental standards.

Notwithstanding the findings, in late 2020, UCIL planned to expand the mine further. This would potentially impact residents of over five villages in the region, who are set to lose their homes and farms. Further, it would heighten the adverse impact on the environment, groundwater and the potential health and livelihoods of the local people. Protests were, therefore, renewed against the possible expansion of the project. In mid-December 2020, several residents of the affected villages held a strike to oppose the expansion of the plant, and simultaneously to demand fair compensation, rehabilitation and employment from the state government, if the expansion were to proceed regardless. The plan to expand the mining project, as well as the opposition to it, is still ongoing.

Lambapur: An Indefinite Suspension of the Uranium Mine

As early as 2003, the intention to establish a mine in the Lambapur-Peddagattu region, in the Nalgonda district, led to 'a sustained protest movement led by various people's organizations' including the Human Rights Forum, Jana Vignana Vedika, and the Progressive Organization of Women and Movement against Uranium Project

(MaUP) (Maheswari, 2003). These organizations were largely civil society groups that were based outside the region, in urban centres such as Hyderabad and Visakhapatnam. The location of the open pit, mines and the purification plant were close to the Rajiv Gandhi Tiger Reserve and the Yellampur sanctuary, as well as the Nagarjuna Sagar reservoir. The adverse impact of radiation and radioactive effluent from the mines and purification plant on the health of the surrounding populace, on groundwater, on the virgin forests, as well as on the water supplied to the Nagarjuna Sagar reservoir, was the major environmental concern. The Nagarjuna Sagar reservoir provided water supply to Hyderabad as well as to various surrounding regions. The initial opposition, therefore, concentrated on the environmental impact of the project, particularly on the protected forest area.

The local people in the region, on the other hand, were initially divided. Several residents of the Peddagattu village, for instance, supported the mine, largely over the possibility of development, jobs, and possible sale of land. Residents of the Lambapur and Seripalle villages, however, opposed the project over fears of displacement, loss of land, and loss of livelihood (*The Times of India*, 2003). Villages such as Seripalle were largely composed of the Lambada, a scheduled tribe community (ST) (indigenous group) whose dependence on traditional modes of livelihood, including small-scale agriculture, is relatively higher. They feared the socio-environmental impact the mine would have on their lives and livelihoods. One resident declared, 'We don't want anything. The water in the tailings pond will be disastrous. Our sheep and animals will die if they touch that water. The groundwater will be contaminated' (Subramanian, Chattopadhyay and Ramu, 2006).

Over the next few years, opposition continued within this region, largely through a collaboration between the residents and the urban-based social organizations. Civil society organizations helped raise awareness about the adverse impact of the uranium project among the local people. MAUP, for instance, states, 'The main objective of the MAUP is to mobilize Public Opinion and to speak in One Voice on major issues' (MAUP, 2005). The tribal people registered their protest through strong non-cooperation. In Seripalle, the populace, along with organizations such as the Jana Vignana Vedika, raised

anti-uranium slogans such as 'We don't want uranium, please save our lives'. In one instance, when personnel from the Atomic Minerals Division (AMD) arrived to explore the region, they were threatened and chased away by the Lambada residents.

In addition to the growing opposition from the local populace as well as the activities of civil society organizations, the presence of Naxalite insurgents in the area presented a further complication. In December 2003, Naxalites kidnapped 26 AMD personnel and set fire to their exploration equipment. They then rounded up the residents of Peddagattu, and 'lectured them on the consequences of uranium mining'. The personnel were freed after four hours (*Frontline*, 2006).

Despite the spirited opposition that was recorded among the local populace, in December 2005, the MoEF gave a conditional environmental clearance to UCIL to set up the uranium mine. Capt. J. Rama Rao filed an appeal against the issuance of environmental clearance with the National Environmental Appellate Authority (NEAA) of the central government. In 2008, however, the NEAA dismissed the case (Ramana, 2013). The only hurdle left was clearance by the Andhra Pradesh state government.

On 3 March 2005, the Andhra Pradesh Pollution Control Board (APPCB) held a public hearing in Lambapur-Peddagattu, as part of the mandatory criteria for granting environmental clearance for the project. It was reported that 'overwhelming opposition was voiced' at the hearing, along with a demonstration at the nearby town of Nalgonda (*Nuclear Monitor*, 2005). Prominent anti-nuclear activists such as Surendra and Sanghamitra Gadekar and Praful Bidwai came down to the hearing to protest the mine and purification plant. One such activist credited the large-scale participation of the local populace, which also included the Lambada community, in the public hearing and the several protest meets that preceded them, for the movement's success. A report by MAUP recounted the protest mounted by the local populace and civil society organizations that day:

The site selected for EPH was the open area on top of a small hillock, with inadequate and shoddy arrangements, and was found to be inconvenient for many of the local people to reach the place. In spite of this, about 1,500

villagers and other concerned citizens and civil society groups at the state and national levels, attended the EPH, under the heavy presence of police force at the site. All through the proceedings, the villagers vociferously expressed their opposition to the project by shouting slogans and exhibiting placards and banners (MAUP, 2005).

A prominent Hyderabad-based social and anti-nuclear activist stated that the public as well as the elected parliamentary members of the constituency the region fell under were in constant opposition to the project. She attributed this opposition to the entrenched leftist sympathies among the people of Nalgonda, and Telangana in general. This included S. Sudhakar Reddy, an MP belonging to the CPI (M) for the Nalgonda constituency, and Ravindra Kumar Ramavath, MLA of the Devarakonda constituency, also of CPI (M). On 20 December 2005, for instance, Sudhakar Reddy spoke in Lok Sabha against the proposed mine, emphasizing its adverse impact on the region's water supply. He said:

The general drainage of the area from all sides is towards Nagarjuna Sagar. Therefore there is every possibility of the radiation-contaminated stormwater runoff from the opencast Uranium mines spread over a vast area, finding its way into Sagar reservoir. This is likely to have serious adverse impacts on the entire downstream Krishna River basin covering about one-third of the total population of the State—Nalgonda, Khammam, Kistna, Guntur and Prakasam Districts—and the food chain in this area, in addition to Twin Cities and part of Ranga Reddy district, which are being supplied drinking water from Nagarjuna Sagar.... Uranium Mining should not be permitted at Lambapur and Peddagattu as it will contaminate Nagarjuna Sagar Reservoir (Reddy, 2005).

The final clearance for the project from the Andhra Pradesh state government, therefore, never came. By its own admittance, this was because a large section of the local tribal communities voiced their strong opposition against the project. Rajeshwar Tiwari, then secretary of the APPCB, clarified that 'More than 50 per cent of those who participated [in the public hearing] opposed the project' (*Frontline*, 2006). It was this opposition, along with the possible adverse impact of the uranium mine and purification plant on the waters of the Nagarjuna Sagar, that convinced the APPCB to refrain from granting clearance for the purification plant.

Consistent political opposition within the state assembly as well as the parliament, combined with consistent social opposition by the local tribal populace of Lambapur-Peddagattu, helped put pressure on the state government to refrain from granting clearance to the uranium mining project. In 2014, N. M. Bahl, a director at UCIL, admitted, 'We could not go ahead with the uranium mining because of opposition by locals and inability of the previous AP government to give a clearance' (*The Times of India*, 2014).

Nallamala: The Campaign to Save the Forest

As mentioned earlier, in 2015, uranium deposits were discovered in the Nallamala forest which led to the proposal to establish another uranium mine in the region. The Nallamala forest, which stretches across several districts in Telangana (particularly the Nalgonda district) and Andhra Pradesh (particularly the Kurnool district), is home to a large population (around 12,000) of the Chenchu, an indigenous group, as well as the Amrabad and Nagarjunsagar-Srisailem tiger reserves, in addition to being an ecologically-sensitive forest cover. The possible socio-environmental impact of a uranium mine caused considerable apprehension to the resident tribal communities. At the same time, the reports on the socio-environmental impact of the uranium mines in Tummalapalle further galvanized opposition against the uranium mine in Nallamala. In June 2019, reports that AMD was granted permission by the central government to conduct exploration in the region led to considerable consternation.

Opposition emerged in both Andhra Pradesh and Telangana states. The Telangana Vidyavanthula Vedika, a local student body, staged a protest against the presence of personnel from UCIL in Nalgonda, and successfully prevented them from entering the Nallamala for initial exploration. At the same time, Jal Biradari, an environmental organization, sent petitions to the chief ministers of Andhra Pradesh and Telangana as well as to the Indian prime minister to rescind the proposed project. Members of the Chenchu tribe also organized themselves in protest: they twice prevented the entry of AMD personnel into the Nallamala to carry out exploration activities. A Chenchu tribal activist stated, 'All of us strongly oppose the proposed mining. It will

destroy this pristine forest and also affect us a lot by way of water contamination... we won't allow anyone to enter' (Janyala, 2019).

This opposition, across several regions in Telangana and Andhra Pradesh, culminated in the 'Nallamala Bandh', a concerted movement involving mass strikes and blockades, undertaken by various organizations and social groups. In the Nagarkurnool district, Telangana, a region abutting the Nallamala forest, activists from among the Chenchu tribe and from other villages organized a series of protests in September 2019. These protests included road blockades through mass sit-ins, mass rallies, as well as the closure of schools, banks and commercial establishments. At the same time, protests also emerged in the Kurnool district, Andhra Pradesh, another region abutting the Nallamala forest. Villagers of the Yadavada Mandal, along with representatives of opposition parties such as the TDP and CPI (M), halted an ongoing exploration by AMD for uranium reserves.

Other civil society organizations, academics and activists, including the Human Rights Forum (HRF), the Hyderabad Tiger Conservation Society, and the prominent leftist writers' association, VIRASAM, joined the burgeoning protests. They held meetings and rallies in cities such as Visakhapatnam and Hyderabad to register their opposition. In all, over 60 social organizations joined the protests to save the Nallamala.

What made the campaign against the uranium mine in Nallamala unique was, however, the successful utilization of social media as a tool of social mobilization. Urban-based activists began a social-media campaign, the 'Save Nallamala Forest' campaign. High-profile celebrities in Telangana and Andhra Pradesh took to Twitter, to register their opposition. This included several prominent actors from the Telugu film industry. Their participation further ensured that the protest resonated among their extensive fan base, including large sections of the rural and urban youth. For instance, Vijay Deverakonda, a popular actor tweeted in support of the 'Save Nallamala' trend on 12 September 2019. His tweet read, 'We have destroyed our lakes, flooded our states, caused drought in others, polluted most of our drinking water sources... Next in line are the lush green Nallamala forests... Uranium

konukovachu – Forests *konukocha*? [Uranium can be bought. Can forests be bought?]. This tweet, highlighting the potentially devastating environmental impact of uranium mining, was 'liked' over 18,000 times, and 'retweeted' nearly 4,000 times.

Pawan Kalyan, a prominent actor-turned-politician and the head of the Jana Sena Party, tweeted out a Telugu translation of the response that Chief Seattle of the Duwamish tribe reportedly gave to the President of the United States in 1852 when the US government sought to acquire the tribe's lands (Kalyan, 2019). The response begins: 'How can you buy or sell the sky, the warmth of the land? The idea is strange to us. If we do not own the freshness of the air and the sparkle of the water, how can you buy them from us?' Pawan Kalyan's tweet centred on the significance of the forests to the tribes that inhabited them. In the case of Nallamala, these were the Chenchu people. The tweet was 'liked' over 16,000 times and 'retweeted' nearly 6,000 times.

Social media also allowed the Chenchu tribe to disseminate their opposition to a large audience. In a protest meeting organized by the Jana Sena Party, Mallikarjun, a prominent Chenchu activist, spoke out against the Indian state's directives on uranium mining: 'Why would you threaten us, who revere the forest and the wildlife that dwell in it as our gods, with a ticking bomb such as this?' (Kalyan, 2019). His speech emphasized the unity of the Chenchu people in opposing uranium exploration and mining in the Nallamala. It further stated that the democratic decision-making process of the Chenchu tribe, which was guaranteed by the Indian constitution, be given precedence over a top-down approach, which the Indian state typically relied on for its nuclear (and developmental) policies: 'As Chenchus, how should we survive, when our right to speak is also being taken away... We make decisions through our PESA gram sabhas (constitutionally-enacted tribal village councils). I ask the government to respect this decision-making process. Do not override them and, without our own knowledge, destroy us.' His impassioned speech was 'liked' nearly 10,000 times on Twitter.

In late September 2019, several activists and organizations belonging to Dalit, Bahujan and Adivasi communities organized the annual

Bathukamma celebrations, a prominent Telangana-specific religious festival that celebrated the mother goddess and nature, around the theme of protecting the Nallamala. On that occasion, the prominent Telangana-based folk singer and activist, Vimalakka, released a song in support of the Save Nallamala campaign. The song was shared extensively on social media platforms, including Twitter. The lyrics, in rough translation, state: 'We shall never allow the footprints of the Chenchu tribe to fade... We shall never allow the multitudes of Mother Forest to be erased...' (Kalyan, 2019).

The Democratic Youth Federation of India (DYFI), the youth wing associated with the CPI(M), began an online petition on Change.org, addressed to the President of India, seeking the halt of the uranium project. The petition garnered upwards of 2,00,000 signatures. The petition read:

Proposed to cover 20,500 acres, the project seems poised to destroy the ecology of the entire [Nagarjunsagar-Srisailem] tiger reserve. The exploration will expose and pollute surface water, groundwater and leach the minerals and dangerous chemicals into the Nagarjunsagar dam. The roads will fragment and degrade the dry forests, which may never recover after such a massive exercise.... The proposal to mine for uranium in this Eden will not only kill its wild denizens but will also take away the livelihoods of the Chenchu, besides exposing them and hundreds of others to uranium contamination (DyFi, 2019).

Following a growing chorus of societal opposition, from among the Chenchu tribal community, from the larger society, as well as opposition political parties, the Telangana state government publicly clarified its opposition to the uranium mining project. On 16 September 2019, the Telangana chief minister, K. Chandrasekhara Rao, categorically stated, 'As far as uranium mining is concerned, following the formation of the Telangana state, no permission was given to anyone, and there is no such plan to give it in the future as well. No one needs to fear about it as at any cost, we will not allow the destruction of Nallamala' (*The News Minute*, 2019). The Andhra Pradesh state government has also not granted its clearance for the establishment of a uranium mine in Nallamala. Finally, with little choice in the matter, the central government followed suit. In late-2019, it rescinded the

permission accorded to AMD to carry out exploration activities in the Nallamala forest.

Purushottam Reddy, professor at Osmania University and a key participant in protests against uranium mining, credited the extensive use of social media (what he referred to as the 'technology of protest') for the Nallamala movement's success. The role and impact of social media in protest movements remain to be understood fully. This is particularly true of protest movements in India. The role of Facebook, as a means of augmenting social mobilization, came to the fore during the Arab Spring in 2011–12. In recent years, in the US and Europe, in particular, the Black Lives Matter campaign relied extensively on social media platforms, such as Twitter, along with on-the-ground mass mobilization. The campaign against the uranium mine in Nallamala is one example of a protest that successfully incorporated social media as a tool of mobilization in India.

FUTURE DIRECTIONS

The examples of Andhra Pradesh and Telangana demonstrate the importance of civil nuclear development in the Indian state. They also provide a glimpse into the future of civil nuclear development, highlighting the importance of commissioning nuclear power plants and uranium (and potentially, thorium) mining, as its twin facets. Uranium mining in India is largely based in Jharkhand, in central India, far away from the nuclear power plants, which are largely located along the Indian coastline. Andhra Pradesh and Telangana are unique in that these regions have demonstrated suitability for the establishment of both nuclear power plants as well as uranium mining projects.

India's nuclear establishment deems the geological terrain of the coastal region of Andhra Pradesh uniquely suitable for a civil nuclear power plant.⁴ A nuclear power plant requires an extensive water supply, which the long coastline can provide. However, as in the case of the proposed uranium mines, the nuclear project in Kovvada has had to face considerable protests from the local populace. As we have seen earlier in this chapter, the opposition movement against the Kovvada nuclear power plant led to considerable delay in land

acquisition for the project. While land was finally acquired, the project itself remains in abeyance. As such, the opposition movement is also dormant. However, if the project is revived in future, the Indian state may well be faced with a revival of the protest movement as well. The population in these regions continue to be concerned about the socio-environmental impact of the nuclear project, and what this would mean for their livelihoods.

The presence of large reservoirs of uranium, perhaps among the largest in the world, means that Andhra Pradesh and Telangana are key to India's civil nuclear ambitions. The Indian government believes that utilizing these uranium reserves will enable the country to rely more extensively on nuclear power as a major source of energy. It will also obviate imports of uranium and help India become more energy independent in the development of nuclear power. However, nearly all of the uranium reserves are located in natural reserves, tiger sanctuaries, close to critical water bodies and reservoirs that supply water to major cities, and the traditional homes of marginalized tribal communities, the Lambadas and Chenchus. As a result, the social, economic and health risks of utilizing uranium in these regions are immense.

The Indian state's attempts to mine the uranium reserves in Andhra Pradesh and Telangana have largely met with failure. The large-scale opposition to uranium mining, since the mid-2000s, has ensured that the extensive reserves in the Nallamala region, including the ones in Lambapur, remain untouched. While the reserves at Tummalapalle are being exploited to a limited extent, plans for its expansion continue to be resisted. The protests against uranium mining brought together diverse sections of the population: tribal communities, activists, NGOs and academia, popular actors as well as politicians. Increasingly, the internet and social media have played a major role in bringing together such diverse opposition. The fact that the 'Save Nallamala' project emerged as a cause celebre in Andhra Pradesh and Telangana played a major part in its success.

The protest movements against nuclear projects in Kovvada, Lambapur-Peddagattu, Nallamala and Tummalapalle together constitute a near-constant opposition against nuclear power, spanning

over two decades, in various parts of Andhra Pradesh and Telangana. As a result, opposition to nuclear energy, and in particular, uranium mining, has become more ingrained within the society and its popular imagination in the two states. For example, in late 2021, a Telugu-language film titled *Akhanda* was released to major commercial success in Andhra Pradesh and Telangana. A central plot of the film involves the hero (played by the star Telugu actor, Balakrishna) thwarting an illegal attempt to mine uranium in his village. In the film, uranium mining is portrayed as being perilous to the surrounding population, who become afflicted with various life-threatening ailments. To the Telugu-speaking people in Andhra Pradesh and Telangana, commercial cinema has always been the most important mode of socio-political communication. The negative portrayal of uranium mining in popular cinema, and its reception by the masses, is probably indicative of how uranium mining (and nuclear energy development in general) is perceived by the larger society.

All the same, so long as nuclear power remains central to India's energy development aims, Andhra Pradesh and Telangana will continue to attract uranium mining and nuclear power plant projects. For the time being, however, protracted social opposition has thwarted these plans. In the face of increasing and ingrained opposition to nuclear energy within the larger society, the implementation of these projects may continue to prove difficult for the Indian state.

NOTES

1. According to the Atomic Energy Act (1962), the state government is responsible for assisting the central government and nuclear establishment in acquisition of land. See DAE (1962).
2. The trajectory of the Kovvada movement (and to some extent, the movements in opposition to uranium mining) has also been established through fieldwork and extensive interviews with residents of the Kovvada region, anti-nuclear activists in Visakhapatnam, Srikakulam and Hyderabad, and members of India's nuclear establishment, from mid-2016 to early 2022.
3. This quote, and most of the quotes in the chapter have been translated from Telugu by the author.
4. Established from interviews with various members of DAE and NPCIL, May to December 2017.